

Self-Learning: Distributed Transactions, TCL & Performance Tuning DBMS

> **Definition:** Distributed Transactions** execute across multiple independent database nodes connected over a network, managed using the **Two-Phase Commit (2PC)**** protocol.

1. Detailed Technical Explanation

1. Two-Phase Commit (2PC) Protocol

Coordinates atomic commit/abort decisions across multiple distributed node participants.

COORDINATOR

2. Transaction Control Language (TCL) Commands

--sql

-- 1. Stop transaction

-- 2. Modify data

UPDATE

-- 3. Set a rollback checkpoint

SAVEPOINT

UPDATE Accounts SET Balance = Balance + 5000 WHERE Acc_No = 102;

-- 4. Rollback to Savepoint if error occurs

ROLLBACK

-- 5. Commit remaining updates permanently

COMMIT

3. Database Performance Tuning Strategies

1. **Indexing Optimization:**

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4. Quick Recall Flow

2PC ->
Indexe